Ghana MoMo Fraud Shield - Project Report

PROJECT: Ghana MoMo Fraud Shield
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1. THE PROBLEM
Mobile money has revolutionized financial inclusion in Ghana, but fraud is a massive barrier. Traditional rule-based systems fail against organized syndicates using SIM boxes and mule chains.

2. THE SOLUTION: HYBRID AI + GRAPH
I built a real-time risk engine scoring transactions in <50ms using:
- Neo4j Graph Database: Maps 10,000+ transactions to expose hidden relationships.
- Graph AI (Risk Propagation): Auto-discovers fraud clusters without hardcoded rules.
- XGBoost ML Model: Catches behavioral anomalies (unusual times, amounts, new recipients).
- FastAPI: Sub-50ms real-time risk scoring.
- Telegram Webhooks: Instant real-time alerting for blocked transactions.

3. LIVE LINKS
Dashboard: https://dey1413-ghana-momo-fraud-shield-dashboard-0se13q.streamlit.app/
GitHub: https://github.com/Dey1413/ghana-momo-fraud-shield